

INDEPENDENT STUDY PROPOSAL

**physio-placebo: Is the LLM Reading the Signal,
or Is the Verbalizer Doing the Classifying?**

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August 2026

1 Overview

physio-placebo: Is the LLM Reading the Signal, or Is the Verbalizer Doing the Classifying?

It is a surrogate-controlled audit of the fast-growing “LLMs reason over your physiology” literature: take published prompting pipelines (Health-LLM style), run them under strict leave-one-subject-out on WESAD / MAUS / CogWear, then systematically replace the physiological signal with spectrum-matched noise, a wrong subject’s recording, a flatline, or nothing at all, and measure how much of the above-chance margin survives. The sharp part is the second move: separating **verbalizer leakage** (the template says “elevated skin conductance, stress-typical”) from actual signal information, which nobody has done.

2 The falsifiable core question

Does an LLM’s above-chance performance on physiological state classification survive replacing the signal with spectrum-matched noise, another subject’s recording, or nothing at all, and does the feature-verbalized condition keep its advantage once class-suggestive adjectives are stripped from the template — or is the measured performance produced by the prompt scaffolding and the label prior rather than by the physiology?

3 Why it is not generic

Closest work	What it does	What it does not do
Health-LLM and the PPG/HRV LLM-agent line	Verbalizes wearable-derived features into prompts, reports gains over baselines	Reports no signal ablation : no noise-matched, no wrong-subject, no signal-deleted condition. The reported margin is never decomposed into signal vs template.
Jo et al., “Are EEG-to-Text Models Working?” (2405.06459)	Invented the discipline this project imports: showed EEG-to-text results survive pure-noise input, i.e. teacher forcing was doing the work	Applies only to EEG→ text generation . Never touches feature-verbalized LLM reasoning, never touches PPG/EDA/HRV, and has no verbalizer concept because there is no verbalizer in that pipeline.
SensorLM / NormWear-style sensor foundation models	Learn representations from physiological streams, benchmark against task baselines	Benchmark against <i>other models</i> , never against <i>surrogate signals</i> . A model that beats a weak baseline while ignoring the input passes their evaluation.

The unclaimed thing: nobody has measured how much of “the LLM read your physiology” is the adjectives in the template. Strip “elevated” and “stress-typical,” leave the numbers and population percentiles, and see what is left. That single ablation is the paper.

4 Concrete semester plan (15 weeks)

Compute reality check: every condition is scored by reading **logprobs over the label tokens in a single forward pass**, not free generation. At 8B in 4-bit with vLLM batching, the full grid is roughly 20–40 GPU-hours. A 16GB card is sufficient.

Datasets (all downloadable week 1, no gatekeeper): WESAD (UCI, 15 subjects, ECG/EDA/BVP/RESP, binary stress vs non-stress as the standard task), MAUS (IEEE DataPort, 22 subjects, wrist PPG under n-back), CogWear (PhysioNet, 11 subjects, Empatica E4 + Muse S, Stroop vs baseline). Stretch: DD-Database (Dryad), OpenNeuro ds003838.

Models: Qwen3-8B-Instruct and Llama-3.1-8B-Instruct (4-bit); Qwen2.5-VL-7B-Instruct for the plot paradigm; optional GPT-class API as a paid upper bound (<\$50).

Weeks	Work
1	Pull all three datasets. NeuroKit2 extraction frozen and version-pinned: HRV time+frequency, EDA tonic/phasic via cvxEDA, respiration. Build the LOSO harness. Read the Health-LLM released code.
2	Classical floor, locked before any LLM runs: L2 logistic regression and a depth-3 decision tree on the identical features, LOSO, per-subject variance reported. Define chance two ways: majority-class, and an empirical floor from 1000 label permutations.
3	Fidelity anchor: reproduce exactly one Health-LLM headline number using their released code. Report the reproduction gap in the paper regardless of size. OSF pre-registration of SDS, the surrogate ladder, the SDS-validity gate (\$5), and the two directional predictions.
4–5	Paradigm A (raw numeric series in prompt) and Paradigm B (verbalized features) x 2 open models x 3 datasets. Prompt-robustness sweep first: 3 templates x {0-shot, 4-shot}, selected on a held-out subject fold, then frozen before any surrogate is run. Each condition gets its own best prompt so “you prompted it badly” is off the table.
6	Paradigm C (matplotlib plot image → Qwen2.5-VL-7B) on WESAD and MAUS.
7–8	Surrogate ladder , applied identically to every paradigm: (1) Gaussian noise matched to <i>globally pooled</i> mean/variance, (2) IAAFT phase-randomized, (3) subject-permuted with label alignment broken (wrong person, independently sampled condition), (4) flatline, (5) signal deleted, scaffolding only. Plus the information-preservation ledger: run the classical LR on every surrogate to quantify how much class information each surrogate actually retains.

Weeks	Work
9	Verbalizer-leakage ablation — promoted into the core, not a stretch. Three template variants on Paradigm B: (i) original adjectives, (ii) neutralized (raw values + population percentile, no evaluative words), (iii) adversarially mis-signed adjectives (say “low” where the value is high) to test whether the model follows the number or the word. This costs three days of compute and is the intellectual centre of the paper.
10	★ WEEK-10 MINIMUM VIABLE RESULT ★ The SDS report card: 3 paradigms x 2 open models x 3 datasets x 6 signal conditions , macro-F1 under strict LOSO, paired-over-subjects bootstrap CIs (1000 resamples, subject as the resampling unit) and Wilcoxon signed-rank across subjects, with the classical-baseline delta and the verbalizer-ablation panel. This is a complete workshop paper whichever direction it points, and it does not depend on any external party.
11–12	Cross-dataset generalization (train verbalizer on WESAD, evaluate on MAUS/CogWear); stretch datasets DD-Database and ds003838 if the pipeline is clean.
13	QLoRA-tuned condition via Unsloth (~6.6GB peak at 8B/2048 tokens) — does fine-tuning make the model <i>start</i> using the signal? Then GPT-class API upper bound.
14	Writeup. Target: ACH or ICMI for the affect framing; ML4H / GenAI4Health workshop for the audit framing.
15	Release physio-placebo : pip-installable, wraps any HF model + any physiological dataset into an SDS report card, plus a reporting checklist modelled on the EEG-to-text mandatory-noise-baseline recommendation. Email the reproduction script to the original author group before submission.

Stretch goals, in priority order: (i) cross-dataset generalization, (ii) QLoRA condition, (iii) API upper bound, (iv) EEG channel from CogWear/ds003838 as a fourth modality.

Pre-registered predictions (to be falsified): SDS < 0.3 for raw-series and plot paradigms; the feature-verbalized paradigm loses >50% of its margin under adjective neutralization; the mis-signed-adjective condition flips predictions in >40% of items.

5 What could kill it, and the fix

Objection 1: “Your surrogates are not nulls, so your audit is measuring nothing.”

Moment-matched Gaussian noise and IAAFT phase-randomization both **preserve per-channel mean and variance** — and mean HR and tonic EDA level are exactly where most of the class information in this data lives. A surrogate that preserves those is leaking the label, which *deflates* SDS and makes a working model look like a placebo. The label-preserving subject permutation as originally specified has the opposite bug: wrong person but right condition is a real, correctly-labelled physiological signal, so high performance there is not evidence of cheating at all.

Fix, built into the design: (a) moments are matched to the **globally pooled** distribution,

never per-class; (b) subject permutation draws a wrong subject **and** an independently sampled condition, so label alignment is destroyed; (c) the label-preserving permutation is retained as a *separate, differently-named* condition (“population-level vs individual-level information”), reported but never counted as a null; (d) the **information-preservation ledger** runs the classical LR on every surrogate and publishes exactly how much class information each one retains, so no surrogate’s status is asserted, it is measured.

Objection 2: “You built it wrong / your SDS is unstable, so the negative result is your bug.”

Two failure modes. First, prompt quality: any null can be dismissed as bad prompting. Second and more serious, $\text{SDS} = (M_{\text{real}} - M_{\text{surrogate}}) / (M_{\text{real}} - M_{\text{chance}})$ **explodes or inverts when M_{real} is near chance** — precisely the case for the raw-numeric-series paradigm, where the denominator is close to zero and the ratio becomes meaningless noise.

Fix: (a) the week-3 reproduce-before-audit fidelity anchor against released code, with the reproduction gap reported honestly; (b) each condition is evaluated at **its own best prompt**, selected on a held-out fold and frozen before surrogates run; (c) a pre-registered **SDS validity gate** — if a condition’s real-signal macro-F1 does not exceed the label-permutation chance floor by a margin whose bootstrap CI excludes zero, SDS is declared undefined for that cell and only the raw delta with CI is reported. No ratio is ever printed on top of a near-zero denominator; (d) subject is the unit of resampling throughout, so the n=11–22 subject counts are honestly reflected in every interval.

Secondary risk: “everyone already knew LLMs can’t read raw PPG.” The defence is that the published literature claims otherwise and reports gains, and the verbalizer arm has a genuinely non-obvious answer either way. If results come out positive, this becomes the first evidence that LLM physiological reasoning survives surrogate controls — still publishable, which is why week-3 pre-registration matters.

6 The project in one sentence

I’d like to test whether the LLMs that claim to reason over physiology are actually using the signal: take WESAD and MAUS, run the published prompting pipelines, then swap the real recording for phase-randomized noise or another subject’s data and see how much accuracy survives, because if it survives then the LLM layer people are bolting onto sensors like ours is mostly reading the prompt template.

More technically:

I want to build a surrogate-control harness for LLM physiological inference: IAAFT phase-randomization, subject permutation and signal deletion across WESAD, MAUS and CogWear under leave-one-subject-out, with a logistic regression on the same HRV and EDA features as the floor, so we can report how much of a paper’s claimed margin is signal and how much is the verbalizer’s adjectives.

The underlying curiosity:

I keep noticing that every “LLM reads your biosignals” paper turns the features into sentences like “elevated skin conductance, stress-typical” before the model ever sees them, and I really want to find out whether the model is classifying the person or just reading that adjective back to us.

7 A second idea, in brief

A gaze-grounded reward-model audit: does the reward-model margin favour responses that cost more to read, after residualizing out length, word frequency, word length and surprisal, using OASST-ETC’s real fixation data scored by Skywork-Reward-V2-8B, ArmoRM and a Qwen3-8B judge. Week 1 must be a hard go/no-go on actually obtaining the OASST-ETC fixation files, since no substitute corpus exists.